

Arian Moradi Asl

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LinkedIn | GitHub | Portfolio

Education

KTH Royal Institute of Technology

Aug. 2021 – Jun. 2026

M.Sc. in Engineering Physics | Specialization in Theoretical Physics

Stockholm, Sweden

- Master's thesis: High-Energy Neutrino Probes of Memory-Burdened Primordial Black Holes

Technical Skills

Programming: Python, MATLAB, TypeScript, SQL

Modelling & Analysis: Computational physics, mathematical modelling, numerical methods, data analysis

Languages: Swedish (Native), English (Fluent), Farsi (Conversational)

Experience

Research Student Assistant, Medical Imaging

Oct. 2025 – Apr. 2026

KTH Royal Institute of Technology

Stockholm, Sweden

- Reviewed literature on stacked prism lenses for X-ray imaging, focusing on their operating principles and reported performance.
- Reviewed research on phase-contrast X-ray imaging and virtual histology, including propagation-based imaging and grating interferometry, with a focus on imaging parameters, resolution, and contrast.

Teaching Assistant

Mar. 2025 – May 2025

KTH Royal Institute of Technology

Stockholm, Sweden

- Led exercise sessions in Classical Physics, supporting students with problem solving and course concepts.

Academic Tutor

Nov. 2021 – Aug. 2025

Allakando

Stockholm, Sweden

- Tutored high school and university students in mathematics and physics, adapting explanations to individual needs.

Projects

Equipment Reservation System

GitHub | Live Demo

Python, FastAPI, PostgreSQL, React, TypeScript

- Built a Python and TypeScript equipment booking application with PostgreSQL, authentication, and role-based access control.
- Implemented transactions and row-level locking to prevent overlapping bookings and added automated API tests for booking rules and access control.

Uptime Monitoring Service

GitHub | Live Demo

TypeScript, NestJS, PostgreSQL, Prisma, React

- Developed a monitoring service that performs scheduled HTTP checks and records response times, status codes, and check history.
- Implemented request timeouts, private-address blocking, persistent data storage, and automated unit and database-backed tests.